



Otolaryngologic manifestations in granulomatosis with Polyangitis: A systematic review of clinical presentations and renal correlation

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Abstract

Background Granulomatosis with polyangiitis (GPA) frequently involves the upper airway and kidneys. Otolaryngologic manifestations often precede systemic symptoms but are commonly misdiagnosed as chronic rhinosinusitis or infection, delaying treatment and increasing the risk of irreversible renal damage.

Aim This review aimed to characterize otolaryngologic presentations in GPA and evaluate their correlation with renal involvement to improve early recognition and clinical outcomes.

Methodology A systematic review was conducted following PRISMA guidelines. They searched PubMed, Scopus, Embase, Web of Science, Cochrane Library, and Google Scholar up to July 2025. They included case reports and case series that described head and neck manifestations in patients with confirmed GPA. Two reviewers independently screened 784 records, selected 17 studies, and extracted data on clinical features, ANCA status, and renal outcomes. The Murad and JBI tools assessed methodological quality.

Results Nasal crusting, epistaxis, septal perforation, and saddle-nose deformity were the most frequent otolaryngologic findings. Oral lesions, including ulcerations and “strawberry gingivitis,” were less common but highly suggestive. Subglottic stenosis, otitis media with effusion, and orbital inflammation also occurred. Many patients received initial misdiagnoses, leading to treatment delays. Renal involvement, primarily glomerulonephritis, developed in most systemic cases, often concurrent with or shortly after severe ENT disease. Destructive lesions correlated strongly with renal complications. PR3-ANCA positivity ranged from 21 to 100%, with most studies reporting rates above 85%. Histopathologic confirmation was achieved through biopsies of nasal or gingival tissue. Immunosuppressive therapy-controlled disease activity, while surgery managed structural sequelae.

Conclusion Destructive otolaryngologic manifestations predicted renal involvement in GPA. Early biopsy, ANCA testing, and multidisciplinary management improved diagnostic accuracy and prevented organ damage. Future research should develop standardized ENT activity scores and prospective registries.

Keywords Anti-Neutrophil Cytoplasmic Antibody · Glomerulonephritis · Nasal Septum Perforation · Renal Insufficiency · Strawberry Gingivitis · Wegener Granulomatosis

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Introduction

Granulomatosis with polyangiitis (GPA) is a systemic vasculitis characterized by necrotizing granulomatous inflammation affecting small- and medium-sized blood vessels. The disease predominantly targets the respiratory tract and kidneys, but the head and neck region frequently show early involvement, often preceding systemic manifestations [1–3]. Otolaryngologic presentations appear in 70–90% of patients, spanning the sinonasal tract, oral cavity, pharynx, larynx, and ear, with clinical features ranging from mild mucosal changes to destructive lesions [2–4]. Timely recognition of these manifestations is essential because untreated disease can quickly progress to severe renal or pulmonary involvement [1, 5, 6].

Nasal involvement constitutes the most common otolaryngologic feature. Patients present with chronic rhinitis, recurrent epistaxis, septal perforation, or saddle-nose deformity caused by cartilage destruction [2, 3, 7]. Sinonasal crusting, polyps, and recurrent infections may mimic chronic rhinosinusitis, delaying diagnosis [4, 8]. Oral manifestations include ulcerations, gingival hypertrophy, and the characteristic “strawberry gingivitis,” which are highly indicative of GPA but occur infrequently [9]. Subglottic and laryngeal involvement can produce airway obstruction, hoarseness, or stridor, sometimes requiring urgent airway management [2, 8]. Ear involvement manifests as otitis media with effusion, sensorineural hearing loss, and external ear granulomas, with these signs occasionally serving as the first indication of systemic disease in both pediatric and adult populations [10–12]. Orbital and ocular lesions such as proptosis, orbital masses, and scleritis could result in permanent vision loss if not treated promptly [13].

Pediatric-onset GPA demonstrates distinct patterns compared with adult disease. Children more frequently exhibit ear involvement, including chronic otitis media, hearing loss, and mastoiditis, while nasal deformities and extensive sinus disease are less common [10]. Otolaryngologic findings correlate closely with systemic disease activity, particularly renal involvement. Persistent sinonasal inflammation and destructive lesions often precede or coincide with glomerulonephritis, making Ear, Nose, and Throat (ENT) assessment a valuable predictor of systemic progression [1, 5, 6]. Several studies indicate that severe or persistent otolaryngologic manifestations, particularly destructive sinonasal lesions or subglottic stenosis, often correlate with active systemic disease and higher risk of renal involvement [2, 5, 6]. Glomerulonephritis frequently develops concurrently with or shortly after pronounced ENT disease, making head and neck findings a clinically useful predictor of kidney involvement [4, 8]. The presence of PR3-ANCA antibodies in patients with ENT manifestations further strengthens this

link, reflecting a shared immunopathologic pathway affecting both renal and otolaryngologic structures [2, 5].

Diagnosis in such cases relies on a combination of clinical evaluation, serologic testing for anti-neutrophil cytoplasmic antibodies (ANCA), and histopathological confirmation of necrotizing granulomatous inflammation [2, 5]. Otolaryngologists play a critical role in the diagnostic process, as a biopsy of accessible head and neck lesions could easily establish the diagnosis, allowing early initiation of immunosuppressive therapy [2, 3]. Treatment strategies incorporate immunosuppressive regimens, including corticosteroids, cyclophosphamide, and biologics such as rituximab, which effectively control otolaryngologic manifestations while minimizing systemic steroid exposure [14]. Surgical interventions address structural complications, including septal perforations, subglottic stenosis, and severe orbital or sinonasal disease [4, 15].

The current available literature on otolaryngologic manifestations remains fragmented. Many studies consist of retrospective analyses or small case series, limiting understanding of the correlation between specific ENT features and renal outcomes [2, 6, 8]. Studies on paediatric patients is particularly scarce, and early ENT interventions have not been systematically evaluated in relation to long-term systemic disease control [10, 11]. Existing reviews tend to focus on individual organ systems, lacking an integrated synthesis of head and neck manifestations alongside renal involvement. This systematic review aims to consolidate clinical evidence on otolaryngologic manifestations of GPA, detailing presentation patterns, diagnostic challenges, and renal correlation. By analyzing both adult and pediatric populations, the review identifies gaps in early recognition and assesses which ENT findings could predict systemic involvement. Findings from this review are intended to guide more timely diagnosis, targeted management, and improved patient outcomes.

Methodology

Research question and PICO(S) framework

This systematic review aims to answer the question: What are the otolaryngologic manifestations of granulomatosis with polyangiitis, and how do they correlate with renal involvement? It was registered with International Prospective Register of Systematic Reviews (PROSPERO) with PROSPERO ID- CRD420251126565. Using a PICO(S) framework, the **population (P)** includes patients diagnosed with granulomatosis with polyangiitis of any age and gender presenting with head and neck manifestations. The **intervention (I)** involves clinical evaluation and diagnostic workup, including histopathology and serology. The

comparison (C) considers patients with GPA without ENT involvement or with different systemic manifestations. The **outcomes (O)** focus on the prevalence, spectrum, and severity of otolaryngologic presentations and their correlation with renal disease. The **study design (S)** encompasses case reports and case series reporting ENT findings in GPA.

Search strategy

The literature search will follow the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines to ensure transparent and reproducible methods. Electronic databases including PubMed/MEDLINE, Scopus, Web of Science, Embase, Cochrane Library, and Google Scholar were searched. Keywords were applied in combination with Boolean operators “AND”, “OR”, and “NOT” to optimize retrieval. Key terms included “*granulomatosis with polyangiitis*”, “*Wegener’s granulomatosis*”, “*otolaryngologic manifestations*”, “*ENT involvement*”, “*nasal involvement*”, “*oral lesions*”, “*laryngeal involvement*”, “*ear involvement*”, “*renal correlation*”, and “*head and neck manifestations*”. Searches included all available literature from 2015 to June 2025 as shown in Table 1.

Eligibility criteria

Studies were included if they reported clinical otolaryngologic manifestations in patients with confirmed GPA, provided sufficient data on clinical presentation or renal involvement, and were published in English. Inclusion was not restricted by age, gender, or study setting. Case reports and case series were only considered. Exclusion criteria encompassed studies lacking clinical detail on ENT findings, review articles, editorials, conference abstracts without full-text availability, non-English language publications, and studies focusing exclusively on non-otolaryngologic manifestations of GPA.

Selection criteria

Inclusion criteria required confirmed GPA diagnosis using accepted clinical, serologic, or histopathological criteria, explicit reporting of otolaryngologic involvement, and sufficient patient-level or aggregate data for extraction. Exclusion criteria eliminated studies with overlapping data, unclear diagnostic confirmation, insufficient description of ENT involvement, or articles without accessible full text.

Data extraction, synthesis, and quality assessment

17 studies were finally included in the present review after careful screening of 784 papers across various databases

Table 1 Database search strategy with keywords, restrictions, and time frame

Database	Search String/Keywords	Boolean Operators	Language/Study Type	Time Frame
PubMed/MEDLINE	“granulomatosis with polyangiitis” OR “Wegener’s granulomatosis” AND (“otolaryngologic manifestations” OR “ENT involvement” OR “nasal involvement” OR “oral lesions” OR “laryngeal involvement” OR “ear involvement”) AND (“renal correlation” OR “head and neck manifestations”)	AND, OR	No language restriction; original research, case series, case reports	2015–June 2025
Scopus	TITLE-ABS-KEY (“granulomatosis with polyangiitis” OR “Wegener’s granulomatosis”) AND (“ENT involvement” OR “nasal involvement” OR “oral lesions” OR “laryngeal involvement” OR “ear involvement”) AND (“renal correlation” OR “head and neck manifestations”)	AND, OR	No language restriction; original research, case series, case reports	2015–June 2025
Web of Science	TS=(“granulomatosis with polyangiitis” OR “Wegener’s granulomatosis”) AND TS=(“ENT involvement” OR “nasal involvement” OR “oral lesions” OR “laryngeal involvement” OR “ear involvement”) AND TS=(“renal correlation” OR “head and neck manifestations”)	AND, OR	No language restriction; original research, case series, case reports	2015–June 2025
Embase	(granulomatosis with polyangiitis’ OR ‘Wegener’s granulomatosis’) AND (‘otolaryngologic manifestations’ OR ‘ENT involvement’ OR ‘nasal involvement’ OR ‘oral lesions’ OR ‘laryngeal involvement’ OR ‘ear involvement’) AND (‘renal correlation’ OR ‘head and neck manifestations’)	AND, OR	No language restriction; original research, case series, case reports	2015–June 2025
Cochrane Library	“granulomatosis with polyangiitis” OR “Wegener’s granulomatosis” AND “otolaryngologic manifestations”	AND, OR	No language restriction; reviews, trials	2015–June 2025
Google Scholar	“granulomatosis with polyangiitis” OR “Wegener’s granulomatosis” AND “ENT involvement” AND “renal correlation”	AND, OR	No language restriction	2015–June 2025

(PRISMA flowchart -Fig. 1). Two independent reviewers (VP and AA) extracted data from each included study using a standardized form capturing patient demographics, type and frequency of otolaryngologic manifestations, diagnostic methods, renal involvement, treatment, and outcomes. A third reviewer (MM) resolved discrepancies to ensure consistency and accuracy. Data synthesis involved descriptive statistics and qualitative analysis to identify patterns in ENT

involvement and renal correlation. Studies were grouped by anatomical site, age category, and type of renal involvement to facilitate structured comparison.

Risk of bias and quality assessment

Methodological quality and risk of bias were evaluated using the frameworks proposed by Murad et al. and Munn

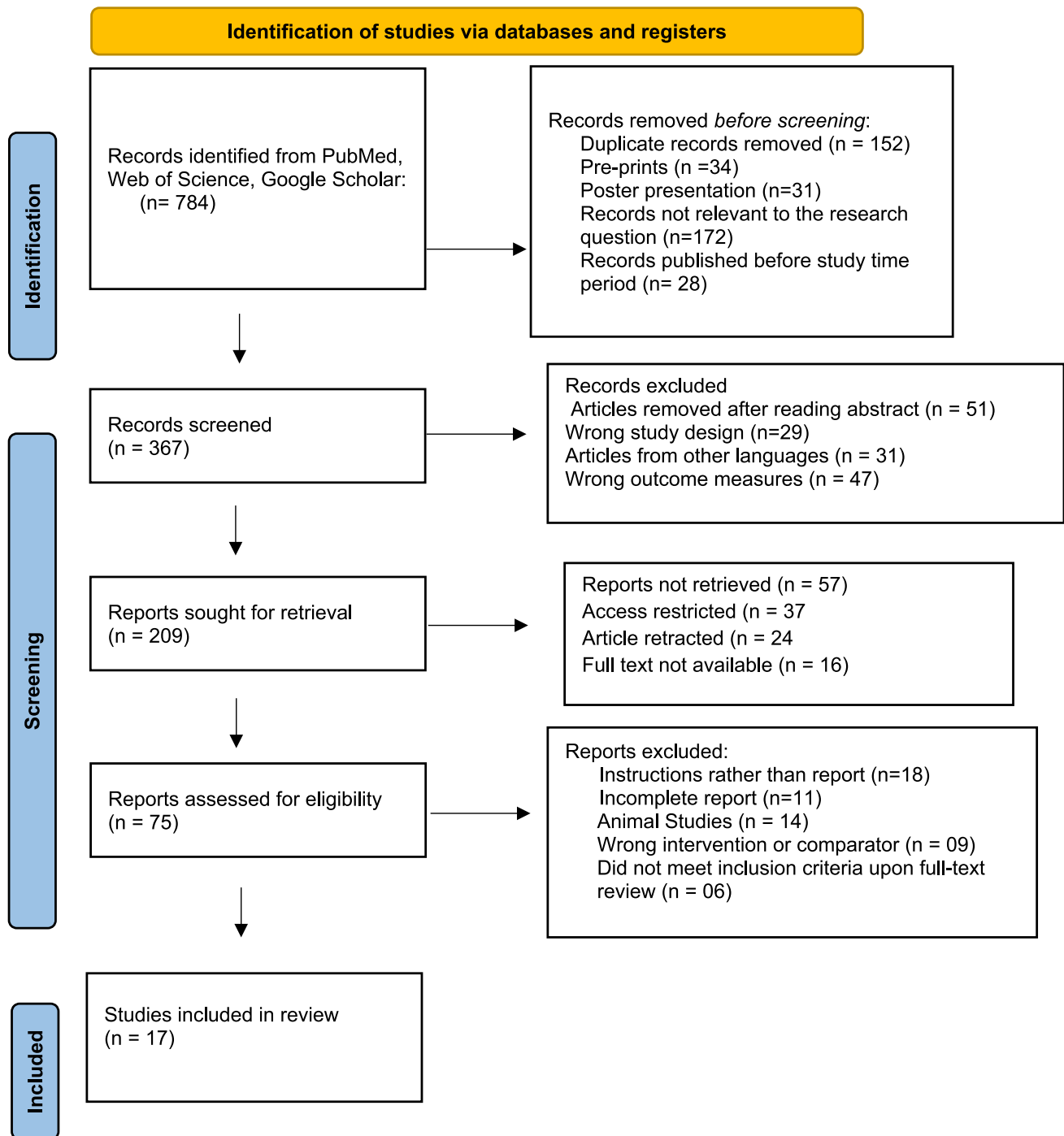


Fig. 1 PRISMA flowchart for the review 2020

et al. Case reports and case series were critically appraised using the Murad et al. tool, which examines selection, ascertainment, causality, and reporting domains [16]. The Joanna Briggs Institute (JBI) critical appraisal tool for case series was applied as outlined by Munn et al., assessing study eligibility, outcome measurement, consecutive patient inclusion, and completeness of follow-up [17]. Each study received an independent evaluation by reviewers VP and AA, with disagreements adjudicated by reviewer MM. Quality assessments informed the interpretation of findings, highlighting limitations related to small sample sizes, and incomplete reporting.

Results

Table 2 summarizes 17 studies, including case reports and case series, describing otolaryngologic manifestations of GPA and their association with renal disease. Reported ENT presentations varied widely, including nasal obstruction, crusting, epistaxis, septal perforation, subglottic stenosis, otitis media with effusion, oral lesions, and “strawberry gingivitis”. Several reports highlighted delayed diagnosis due to misinterpretation as chronic rhinosinusitis, nasal polyposis, or common oral infections [12, 25, 26]. Renal involvement was frequent, particularly in systemic GPA, manifesting as hematuria, proteinuria, and glomerulonephritis, although some patients had ENT-limited disease without renal pathology [6, 10–12, 19, 23, 26]. ANCA serology, especially PR3-ANCA positivity, served as a principal diagnostic marker, and repeated testing proved critical in initially negative cases [11, 12, 21]. Histopathological confirmation from accessible ENT sites such as gingiva or nasal mucosa facilitated early detection [18, 20, 22, 24, 25]. Immunosuppressive therapy, including corticosteroids, cyclophosphamide, methotrexate, and rituximab, was the mainstay of treatment, while surgical interventions were applied for airway obstruction or structural damage [4, 6, 12, 26]. Pediatric cases showed high rates of subglottic stenosis and severe airway complications [10]. These studies emphasize the diagnostic and prognostic significance of ENT manifestations in predicting systemic progression and renal involvement in GPA.

Table 3 presents the quality assessment the case reports using the tool developed by Murad et al. The tool comprises four domains: *selection (representativeness of the case)*, *ascertainment (A1–A2: accurate exposure and outcome measurement)*, *causality (C1–C4: temporal relationship, alternative explanations, dose–response, and challenge–rechallenge)*, and *reporting (clarity and completeness of case description)*. Each item is scored 1 (yes) or 0 (no), with a maximum of 8 points. Scores of 7/8 indicate high

quality, 6/8 moderate-to-high quality, and 5/8 moderate quality. Most studies scored 6 or higher, demonstrating generally reliable methodology and adequate reporting. High scores were largely attributed to precise diagnostic confirmation, detailed clinical timelines, and comprehensive treatment documentation, while lower scores were linked to incomplete follow-up or limited exploration of alternative diagnoses.

Figure 2 summarizes the risk of bias assessment for the included case reports using the JBI Critical Appraisal Checklist, which evaluates 10 domains of methodological quality: *clear reporting of patient demographics, detailed clinical history, presentation, diagnostic methods, treatment details, outcomes, adverse events, timelines, patient perspective, and overall key findings*. Each criterion was scored as “+ +” (fully met), “+” (partially met or unclear), “–” (not met), or “NA” (not applicable). Most studies demonstrated high reporting quality with several “+ +” ratings, including Sahyouni et al. (2019) [11] and Szczklik et al. (2019) [23], while studies like Sung et al. (2015) [18] and Hanisch et al. (2016) [19] had occasional “+” or “–” due to incomplete timelines or adverse event reporting. A few criteria were marked “NA” where not relevant, for example, Amiraraghi et al. (2015) [12] and Wojciechowska et al. (2018) [8]. Hence, the studies done by Felicetti et al. (2018) [6], Maqbool et al. (2023) [26], and Górny et al. (2025) [10] constantly scored “+ +” in most domains, depicting low risk of bias and strong methodological rigor, which was similar to Table 2 where the majority of reports scored between 6/8 and 7/8. This indicated moderate-high to high quality and thus supports the credibility of their findings.

Figure 3 illustrates the proportion of ANCA-positive cases reported in each included study, showing distinguished variation across articles. Amiraraghi et al. (2015) [12], Sung et al. (2015) [18], Hanisch et al. (2016) [19], Fonseca et al. (2021) [20], Felicetti et al. (2018) [6], Szczklik et al. (2019) [23], and Dhalkari et al. (2020) [24] each documented a full 100% ANCA positivity rate, indicating all patients tested positive for ANCA in these case reports. Wojciechowska et al. (2018) reported 89% positivity, while Patrick et al. (2019) [22] and Sahyouni et al. (2019) [11] also demonstrated consistently high values of 100%. Carnevale et al. (2019) [4] observed 88% positivity, and Nico et al. (2020) [25] reported a slightly lower 75%. Maqbool et al. (2023) again showed a 100% positivity rate, whereas Górny et al. (2025) [10] reported 84.6%. The only study with a substantially low proportion was D’Onofrio et al. (2023) [27], with 21% ANCA positivity, highlighting variability in serological profiles among cases. These findings show both heterogeneity in patient populations and differences in study sizes, with most studies recording high ANCA positivity thus strengthening its strong diagnostic association of the disease.

Table 2 Summary of case reports and case series describing ENT manifestations in granulomatosis with polyangiitis, highlighting clinical features, ANCA status, renal involvement, treatment, and outcomes

SI no	Author/Year	Study Design	Country	Sample Size & Demographics	ENT Manifestations	ANCA Status	Diagnostic Method	Renal Involvement	Treatment	Follow-up	Key Outcomes
	Amirraghi et al., 2015 [12]	Case series	UK (NHS Lanarkshire & NHS Greater Glasgow, Scotland)	4 patients (3 adults: 58 M, 64 M, 47 F; 1 pediatric: 15 F)	Otitis media with effusion, worsening post-myringotomy; acute ear infection; mastoiditis; external canal granulations; bilateral hearing loss; bilateral tympanic membrane granulations; nasal obstruction; severe right and left LMN facial nerve palsy; facial pyoderma-like lesions	Proteinase 3 (PR3) ANCA positive in all cases; initial ANCA negative in 2 cases but positive on repeat	Histopathology (renal biopsy in case 1: GPA confirmed; nasal mucosa granulations); ANCA serology; temporal bone CT; nasal endoscopy; ENT biopsy	Case 1 had renal biopsy confirmed GPA; others diagnosed before systemic renal disease developed	Immunosuppressants + corticosteroids (specific agents not detailed); surgical interventions (grommet insertion, mastoidectomy, biopsies)	Serial follow-up (3–6 months intervals recommended; duration not specified)	Primary otologic GPA presentations are rare; delayed ANCA positivity; early clinical suspicion critical to prevent systemic disease; surgical interventions may worsen symptoms; repeat ANCA testing recommended
	Sung JY et al. (2015) [18]	Case Report	Korea	1 patient; 57-year-old male sailor	Gingival swelling with "strawberry-like" appearance; orofacial granulomatous lesions; facial skin ulcers	Positive cytoplasmic ANCA (c-ANCA); elevated PR3-ANCA levels	Clinical presentation, histopathological biopsy (showing chronic active inflammation, micro-abscesses, fibrinoid vasculitis, palisading granulomas), and positive c-ANCA/PR3-ANCA testing	No renal involvement (normal renal function and urinary sediment tests)	Methotrexate 12.5 mg once weekly; Prednisone 40 mg daily (tapered to 2.5 mg for maintenance over 2 months)	2 months	Rapid improvement after 10 days; no relapse of facial or gingival lesions at 2-month follow-up; alveolar bone re-covered with mucosa post-treatment
	Hanisch M et al. (2016) [19]	Case report	Germany	1 patient; 72-year-old female	Hyperplastic gingivitis (classical "strawberry gingivitis"), intraoral pain, oral ulcerations	c-ANCA positive	Clinical exam, histopathology (gingival biopsy), ANCA serology	Glomerulonephritis, nephrosclerosis	Prednisolone, Cyclosporine, Rituximab (375 mg/m ² weekly × 4 doses)	Resolved oral lesions; clinical remission achieved	Oral findings led to GPA relapse diagnosis; rituximab effective; emphasizes oral mucosa as diagnostic site

Table 2 (continued)

SI no	Author/Year	Study Design	Country	Sample Size & Demographics	ENT Manifestations	ANCA Status	Diagnostic Method	Renal Involvement	Treatment	Follow-up	Key Outcomes
	Fonseca FP (2021) [20]	Case Report	Brazil	1 patient; 75-year-old female with medical history of osteoporosis, arthrosis, and hypothyroidism	Gingival swelling (erythematous hyperplasia with granular surface), predominantly affecting the lower anterior gingiva from right second premolar to left second premolar; "strawberry-like" gingivitis appearance; frequent bleeding; no ulceration or tooth mobility	Positive c-ANCA (by indirect immunofluorescence); Positive PR3-ANCA (by ELISA)	Clinical presentation, incisional biopsy (showing mixed inflammatory infiltrate, microabscesses, multinucleated giant cells, pseudoeplitheliomatous hyperplasia), immunohistochemistry (CD3, CD20, CD68, CD138), negative histochemical stains (PAS, Ziehl–Neelsen, Grocott), chest CT (showing diffuse lung opacification), lung biopsy (confirming granulomatous reaction)	No renal involvement (renal tests were normal)	Methylprednisolone 40 mg/day; Azathioprine 50 mg every 12 h	3 months (until death)	Patient developed constitutional symptoms (dyspnea, fever, diarrhea, vomiting, weight loss, sepsis, submandibular lymphadenopathy) 2 weeks after diagnosis; developed CMV infection requiring interruption of immunosuppression; despite restarting treatment, progressed to chronic anemia, cardiac insufficiency, pulmonary hypertension, and uncontrolled systemic infections; died 3 months after diagnosis
	Kühn et al., 2018 [7]	Large retrospective case series (1990–2012)	Germany (Christian-Albrechts-University of Kiel)	230 GPA patients; demographics not specified in full; >95% with ENT involvement	Nasal obstruction (59%), loss of smell (57%), nasal crusting (51%), bloody crusts (33%), purulent crusts (19%), saddle nose deformity (24%), septal perforation (29%), orbital involvement, otitis media, hearing loss; ENT involvement in >95%	c-ANCA/PR3-ANCA predominant (exact numbers not given); p-ANCA rare	Histopathology, ANCA serology (IIF, ELISA), ENT endoscopy, CT/MRI imaging, ENTAS scoring	Renal involvement not quantified; systemic disease noted, ENT disease often precedes renal	Immunosuppressive therapy (cyclophosphamide, corticosteroids, methotrexate, rituximab); surgeries for severe nasal/airway damage	Long-term; ENTAS scoring for follow-up, recurrent assessments	ENT disease was earliest sign in majority; >95% ENT involvement; high diagnostic yield of ENT findings; ENTAS effective in monitoring

Table 2 (continued)

Sl no	Author/Year	Study Design	Country	Sample Size & Demographics	ENT Manifestations	ANCA Status	Diagnostic Method	Renal Involvement	Treatment	Follow-up	Key Outcomes
	Wojciechowska et al., 2018 [8]	Comparative case series (GPA vs. MPA)	Poland (Wrocław Medical University)	64 patients (41 GPA, 23 MPA); mean age GPA 50.2 yrs, MPA 56.2 yrs; 31 females, 33 males	ENT involvement in 75% overall; chronic rhinosinusitis, epistaxis most common; symptoms preceded systemic disease by mean 14.4 months	57/64 ANCA positive (89%); 34 PR3-ANCA (59.6%), 21 MPO-ANCA (36.8%), 7 ANCA-negative	ANCA serology (ELISA, IIF), histopathology, ENT clinical evaluation, imaging	Renal involvement in 73.44% overall; more frequent in MPA	Immunosuppressants (cyclophosphamide, corticosteroids); individualized therapy	Not specified (retrospective records)	ENT symptoms often first sign of systemic vasculitis; GPA associated with pulmonary/ENT dominance; renal disease more frequent in MPA
	Felicetti et al., 2018 [6]	Retrospective cohort study (chart review, 2005–2015)	Italy (University of Verona)	112 GPA patients (52 men, 60 women); median age 54 yrs (range 16–85)	Sinonasal (86.6%), ear (37.5%), subglottic stenosis (15.2%), orbital (13.4%), laryngeal (9.8%); ENT symptoms first in 54%	97.3% ANCA-positive; 90.2% PR3-ANCA, 7.1% MPO-ANCA	ANCA serology, histopathology (ENT biopsies), ENT exams, CT/MRI imaging	63.4% renal involvement at diagnosis; higher rate in systemic phenotype	Immunosuppressive therapy (cyclophosphamide, methotrexate, rituximab, corticosteroids); ENT surgeries for stenosis, deformities	Median follow-up: 8.2 yrs (IQR 5.4–11.7)	ENT disease correlated with systemic severity; ENT involvement at onset predicted higher risk of renal damage; subglottic stenosis prevalent

Table 2 (continued)

ISI no	Author/Year	Study Design	Country	Sample Size & Demographics	ENT Manifestations	ANCA Status	Diagnostic Method	Renal Involvement	Treatment	Follow-up	Key Outcomes
	Thompson G et al. (2018) [21]	Case Series (2 cases)	Australia	Case 1: 43-year-old male Case 2: 63-year-old female with prior history of polymyalgia rheumatica	Case 1: Strawberry gingivitis (erythematous gingivae with petechiae, ulceration of all gingival mucosa) Case 2: Localized hyperplastic gingival swelling on upper labial gingiva (teeth 14–24), vascular, "strawberry-like" appearance	Case 1: Negative c-ANCA; negative anti-PR3 antibodies Case 2: Low-positive c-ANCA (3 units, normal < 4); low-positive PR3-ANCA (1.3 U/ml, ref < 1 U/ml)	Clinical presentation (strawberry gingivitis), gingival biopsy (showing pseudoeplitheliomatous hyperplasia, neutrophil microabscesses, inflammatory infiltrate; Case 2: also multinucleated giant cells and one loose granuloma), serology (ANCA/PR3 testing), CT imaging (sinuses, chest)	No renal involvement in either case (no evidence of renal disease on investigations)	Case 1: Initial: oral sulfamethoxazole/trimethoprim + prednisolone 25 mg → escalated to prednisolone 50 mg/day + topical prednisolone liquid + IV cyclophosphamide 1000 mg (CYCLOPS regimen). Maintenance: azathioprine 100 mg + tapering prednisolone. Relapse treated with prednisolone 50 mg + IV rituximab 1 g × 2 doses Case 2: Prednisolone 1 mg/kg → methotrexate added → IV cyclophosphamide (CYCLOPS) for progression	Case 1: Follow-up until relapse at 5 months post-treatment initiation; long-term outcome included significant gum recession low-up until complete resolution after IV cyclophosphamide Case 2: Follow-up until complete resolution after IV cyclophosphamide	Case 1: Initial deterioration despite treatment; improvement after IV cyclophosphamide; relapse after 5 months; ultimately achieved remission with rituximab. Significant periodontal destruction and gum recession remained Case 2: Dramatic improvement with steroids; progression to epistaxis and sinusitis on methotrexate; complete resolution after IV cyclophosphamide

Table 2 (continued)

Sl no	Author/Year	Study Design	Country	Sample Size & Demographics	ENT Manifestations	ANCA Status	Diagnostic Method	Renal Involvement	Treatment	Follow-up	Key Outcomes
	Sahyouni et al., 2019 [11]	Case series, retrospective chart review (2010–2016)	USA (University of California, Irvine)	11 patients (4 men, 7 women); mean age 49.2 years (range 19–78)	Otitis media with effusion (OME) in 10/11; Eustachian tube dysfunction (ETD) in 1; mixed/conductive/sensorineural hearing loss (HL) (91%); unilateral or bilateral HL; otalgia (6/11); facial paralysis (1); intranasal granulation (3); subglottic stenosis (1); gingival lesions in some cases	c-ANCA positive in all patients eventually; 3 patients had negative initial c-ANCA, positive on repeat	Diagnosis per ACR and CHCC criteria; c-ANCA serology; histopathology (gingival/sinonasal biopsies); nasal endoscopy; CT and MRI	2 patients with initially negative c-ANCA had proteinuria or hematuria on urinalysis, indicating renal involvement	Prednisone 1 mg/kg/day (up to 80 mg); methotrexate 0.3 mg/kg/week; adalimumab for maintenance; trimethoprim-sulfamethoxazole for isolated ear disease; PE tube placement	Average of 4 in-office follow-ups; repeat ANCA testing recommended	Otologic GPA may be first or only manifestation; early repeat ANCA testing critical; immunosuppressive; PE tubes relieved OME; facial paralysis improved from grade VI to III
	Carnevale et al., 2019 [4]	Retrospective case analysis	Spain	19 patients; 9 male, 10 female; age 19–68 years (mean 46.47)	Sinonasal involvement (56%), laryngotracheal (31.25%), ear (25%), subglottic stenosis, nasal obstruction, rhinorrhea, septal perforation, saddle nose, sudden SNHL	14/16 ANCA positive: 11 c-ANCA (PR3), 3 p-ANCA (MPO); 2 negative	Histopathology (nasal mucosa biopsy; 5/7 positive), immunofluorescence, CT/MRI	63.1% (12/19) with renal involvement	Immunosuppressants, corticosteroids, balloon dilations, CO ₂ laser resections, cricotracheal resection, rhinoseptoplasty, tympanostomy tubes	Up to 7 years; airway patency maintained post-surgery	ENT symptoms were first sign in 43.75%; nasal biopsy highly diagnostic; surgery effective in severe airway cases

Table 2 (continued)

SI no	Author/Year	Study Design	Country	Sample Size & Demographics	ENT Manifestations	ANCA Status	Diagnostic Method	Renal Involvement	Treatment	Follow-up	Key Outcomes
	Patrick A et al. (2019) [22]	Case report	United Kingdom (UK)	1 patient; 56-year-old non-smoking female with fibromyalgia	Gingival bleeding, swelling, pain in all four quadrants; erythematous and tender 20 × 10 mm lesion in upper right quadrant; speckled appearance of upper right labial gingiva; palatal involvement; dry cough, blocked nose, blood in right nostril	c-ANCA positive (cytoplasmic antineutrophil cytoplasmic antibodies)	Incisional biopsy (showed partially ulcerated non-keratinising squamous epithelium, microabscesses with numerous eosinophils, foreign body-type giant cells, minimal leukocytoclastic vasculitis, Elastic Van Gieson stain showing destruction of elastic vessel wall); blood tests (↑ platelets, ↑ ESR, ↑ gamma-glutamyl transpeptidase, IgG positive for HSV-1 and EBV, IgM negative); nasendoscopy (3–4 mm epithelial changes, small haemorrhagic region, mucosal excoriation, excess secretions); CT head and jaws (normal, no bony erosion); orthopantomogram (OPG) (no bone loss or pathology)	No overt renal failure; adequate renal function noted; haematuria not explicitly mentioned but c-ANCA positivity and systemic features suggest risk	Prednisolone and co-trimoxazole (trimethoprim-sulfamethoxazole); gingival contouring (paring of lesion for cosmetic result)	Continuing follow-up by rheumatology team; lesion reduced in size over time; prognosis described as good	Gingival lesions can be primary presentation of GPA; dentist was first clinician; initial misdiagnosis as acute ulcerative gingivitis; biopsy critical for diagnosis; early diagnosis leads to good prognosis; oral lesions may mimic common gingival conditions; up to 5% of GPA cases present with oral lesions as primary feature; c-ANCA positivity confirms diagnosis; systemic symptoms (cough, nasal bleeding) correlated with ENT findings

Table 2 (continued)

SI no	Author/Year	Study Design	Country	Sample Size & Demographics	ENT Manifestations	ANCA Status	Diagnostic Method	Renal Involvement	Treatment	Follow-up	Key Outcomes
	Szczekliak K et al. (2019) [23]	Prospective cohort study (June 2014 – November 2017)	Poland (Jagiellonian University Medical College, Kraków)	9 patients: 4 females, 5 males; age range 24–68 years (median 57, IQR 50–64); median disease duration 2 years (IQR 1–6 years); all met ACR 1990 criteria for GPA	- Strawberry gingivitis (1 patient, histopathologically confirmed) - Gingivitis (11.1%) - Tongue filiform papillary hypertrophy (22.2%) - Petechiae of oral mucosa (22.2%) - Coated tongue (11.1%) - Bone destruction and inflammatory lesions in paranasal sinuses (55.6% on CBCT) - Maxillary sinus wall destruction, mucosal thickening, bone and soft tissue masses in sinuses (imaging) - No salivary gland enlargement or pain	- ANCA positive in all 9 patients (immunofluorescence method) - Titers ranged from 1:10 to 1:1280 - All had elevated anti-PR3 antibodies (ELISA) - Median anti-PR3: 49 IU/ml (range: <0.4–156 IU/ml) - Median PR3 level: 87 IU/ml (range: 4–156 IU/ml)	- ACR 1990 criteria for GPA diagnosis (≥ 2 criteria required) - Dental examination: external and internal inspection, WHO protocol, mirror, explorer - Histopathological examination of gingival biopsy (confirmed strawberry gingivitis) - Cone Beam Computed Tomography (CBCT) of maxilla and mandible - Microbiological cultures: bacterial and fungal swabs from cheek and tongue - ANCA testing: immunofluorescence and anti-PR3 ELISA - Laboratory tests: blood samples for PR3, ESR etc	- 5 out of 9 patients (55.6%) had kidney involvement - Manifestations: abnormal urinary sediment (RBCs or red cell casts) - One patient had disease exacerbation with renal symptoms	- Induction therapy: high-dose glucocorticosteroids (3 pulses of 250–500 mg methylprednisolone IV, followed by 1 mg/kg/day, then tapering) + cyclophosphamide (in 3 patients) - Maintenance therapy: low-dose glucocorticosteroids + azathioprine (2), methotrexate (1), or rituximab (2) - No local surgical intervention reported for ENT lesions	- Median follow-up not specified; study duration: June 2014 – November 2017 - Patients assessed during single visit for dental/oral evaluation - Longitudinal disease activity monitored: 6 in remission, 1 with exacerbation, 2 newly diagnosed	- Oral lesions present in 55.6% of GPA patients - Strawberry gingivitis rare (11.1%) but highly specific and may be first sign - CBCT detected sinonasal bone destruction - In 55.6%, highlighting value of dental imaging - Fungal infections (Candida) in 66.7% (5C, <i>albicans</i>, 1C, <i>norvegensis</i>), likely due to immunosuppression, 2 newly diagnosed - No cartilaginous bacteria found; all patients had dental caries (mean DMF-T = 22) - Emphasizes interdisciplinary care: dentists must refer patients with atypical gingivitis or radiological bone loss for systemic evaluation - Early diagnosis via oral findings can prevent irreversible organ damage

Table 2 (continued)

Sl no	Author/Year	Study Design	Country	Sample Size & Demographics	ENT Manifestations	ANCA Status	Diagnostic Method	Renal Involvement	Treatment	Follow-up	Key Outcomes
	Dhalkari CD et al. (2020) [24]	Case report	India (Government Dental College and Hospital, Aurangabad, Maharashtra)	1 patient; 17-year-old male	Localized gingival enlargement in maxillary anterior region (from distal of right first premolar to distal of left first premolar); erythematous, friable, painful, and bleeding gingiva resembling "strawberry gingivitis"; no other oral lesions; no nasal or laryngeal symptoms reported	Antineutrophil cytoplasmic autoantibody (ANCA) titer:1:23 (positive)—described as c-ANCA (implied by context and positivity)	Clinical examination: "strawberry gingivitis" appearance -Incisional biopsy under local anesthesia -Histopathology: parakeratinized stratified squamous epithelium, pseudoepitheliomatous hyperplasia, granulomatous inflammation, mixed infiltrate (neutrophils, eosinophils, plasma cells, lymphocytes), multinucleated giant cells, pronounced vasculitis -Special stains: PAS and Grocott-Gomori methenamine-silver nitrate (negative for fungal infection) -Systemic workup: chest radiographs (normal), Mantoux test (negative), peripheral blood smear (normal), hemoglobin, CBC, ESR, urine analysis (all normal)	No renal involvement reported;urine analysis normal; no hematuria, proteinuria, or red cell casts; no clinical or laboratory signs of glomerulonephritis	Prednisolone: 20 mg/day, tapered over time	1-week follow-up: improvement noted - 3-week follow-up: continued improvement - Additional 3 months of follow-up: stable, lesion-free, no recurrence	Strawberry gingivitis was the first and only presenting sign of WG - Diagnosis confirmed by clinical, histopathological, and serological correlation - No systemic organ involvement at diagnosis (limited form) - Rapid response to corticosteroids - Emphasizes dentist's critical role in early diagnosis - Differential diagnoses (fungal infection, TB, leukemia, lymphoma) were ruled out systematically - Case fulfilled 2 ACR 1990 criteria: (1) oral ulceration ("strawberry gingivitis"), (2) granulomatous inflammation on biopsy

Table 2 (continued)

SI no	Author/Year	Study Design	Country	Sample Size & Demographics	ENT Manifestations	ANCA Status	Diagnostic Method	Renal Involvement	Treatment	Follow-up	Key Outcomes
Nico MMS et al. (2020) [25]		Case Series (4 cases)	Brazil	Case 1: 45-year-old female	Case 1: "Strawberry gingivitis" – hemorrhagic, vegetating mass on lower gingiva; teeth mobility; alveolar bone destruction	Case 1: Negative ANCA Case 2: c-ANCA 1:320 Case 3: c-ANCA 1:160 Case 4: c-ANCA 1:160	Clinical presentation, histopathological biopsy (showing pseudoeplitheliomatous hyperplasia, microabscesses, necrosis of vessels, perivascular granulomas), serology (c-ANCA), imaging (CT for sinusitis in Case 4), exclusion of infections (negative stains for microorganisms), and application of Chapel Hill 2012 criteria	Case 1: No renal involvement	All patients received prolonged cyclophosphamide as per standard guidelines	Case 1: 16 years (no relapse after initial mild relapse) Case 2: 9 years (asymptomatic) Case 3: Lost to follow-up Case 4: 6 months (no relapse)	Case 1: Complete clearance after treatment; mild relapse observed; long-term remission; confirms ANCA-negative GPA can present with classic strawberry gingivitis Case 2: Rapid progression of oral lesion paralleled severe systemic disease; full recovery with treatment Case 3: Coexistence with Crohn's disease; severe oral and pulmonary involvement; lost to follow-up Case 4: Diagnosis delayed due to initial misdiagnosis as leishmaniasis; palatal perforation persisted despite treatment; highlights mimicry by endemic infections in Brazil
				Case 2: 43-year-old female				Case 3: No specific mention of renal disease			
				Case 3: 35-year-old male				Case 4: No renal involvement			
				Case 4: 62-year-old female							
Maqbool et al., 2023 [26]		Case report	Pakistan	1 patient; 32-year-old male	Nasal obstruction, recurrent epistaxis, septal perforation, saddle-nose deformity, nasal crusting, sinonasal polypsis (initial misdiagnosis)	c-ANCA > 200 U/mL (positive)	Histopathology: chronic necrotizing granulomatous inflammation; c-ANCA serology; HRCT chest (bilateral consolidation, ground-glass opacities); ENT exam	Proteinuria +1, hemoglobin +3 on urine analysis; glomerulonephritis confirmed	Methotrexate 15 mg weekly + folic acid; Prednisolone 5 mg/kg/day tapered; TMP-SMX (Septra); calcium & oneprazole	2-weekly follow-up (duration not specified)	Delayed diagnosis due to misinterpretation as nasal polyposis; ENT and pulmonary correlation essential; successful initiation of immunosuppression

Table 2 (continued)

Sl no	Author/Year	Study Design	Country	Sample Size & Demographics	ENT Manifestations	ANCA Status	Diagnostic Method	Renal Involvement	Treatment	Follow-up	Key Outcomes
	D'Onofrio M et al. (2023) [27]	Case series (cohort of EGPA patients with ENT disease; comparative literature review)	Italy	14 patients (7 M, 7 F); mean age 54.1 yrs (range 27–74); retrospective study	Nasal obstruction (93%), nasal polyps (86%), anosmia/hyposmia (64%), chronic rhinosinusitis, otitis media (36%), hearing loss (43%), facial pain/headache, orbital involvement rare	ANCA-positive in 3/14 (21%); MPO-ANCA in 2; PR3-ANCA in 1	ANCA serology (IIF, ELISA), histopathology, nasal endoscopy, CT/MRI imaging	5/14 patients (35.7%) had renal involvement: microscopic hematuria, proteinuria, mild GN	Systemic corticosteroids, mepolizumab, cyclophosphamide, azathioprine, surgery (functional endoscopic sinus surgery, PE tubes)	Mean follow-up: 6.5 yrs (range 1–10)	ENT disease frequent and severe; nasal polyps hallmark; EGPA patients often ANCA-negative; renal involvement less severe than GPA; mepolizumab reduced ENT exacerbations
	Górny et al., 2025 [10]	Case series	Poland	15 pediatric patients (9 girls, 6 boys); age 8–17 yrs; mean 14.1 yrs	Nasal obstruction, epistaxis, saddle nose, sinusitis, septal perforation, nasal discharge, vestibular pain, orbital pseudotumor, subglottic stenosis (10/15), tracheal stenosis, bronchoconstriction, stridor, hoarseness, respiratory failure, ear pain, otitis media, hearing loss, tinnitus, CN VII palsy	c-ANCA positive in 11/13; PR3-ANCA positive in 4; MPO-ANCA positive in 1	Histopathology (necrotizing granulomatous vasculitis, focal segmental glomerulosclerosis), imaging (CT), ANCA serology	7 patients: proteinuria, hematuria, necrotizing glomerulonephritis	Corticosteroids, cyclophosphamide, rituximab, other immunosuppressives; airway interventions for stenosis	Variable; some deaths due to airway obstruction; others improved with immunosuppressive therapy	Pediatric GPA shows ENT-predominant onset; subglottic stenosis frequent; systemic progression with kidney and lung disease

Footnote: ANCA: anti-neutrophil cytoplasmic antibody; PR3: proteinase 3; MPO: myeloperoxidase; c-ANCA: cytoplasmic ANCA; p-ANCA: perinuclear ANCA; ENT: ear, nose, throat; GPA: granulomatosis with polyangiitis; EGPA: eosinophilic granulomatosis with polyangiitis; GN: glomerulonephritis; CT: computed tomography; MRI: magnetic resonance imaging; CBCT: cone beam computed tomography; OME: otitis media with effusion; HL: hearing loss; PE: pressure equalization; TMP-SMX: trimethoprim-sulfamethoxazole; IV: intravenous; ESR: erythrocyte sedimentation rate.

Discussion

This review confirmed that ENT involvement in GPA is common and often an early sign of systemic disease, especially renal complications. Sinonasal disease was reported in 70–90% of patients and frequently preceded systemic diagnosis by several months [3, 28]. Carnevale et al. documented septal perforation in 47% of 19 cases, showing its diagnostic relevance [4]. More than half of the studies reported an initial misdiagnosis as chronic rhinosinusitis or nasal polyposis, delaying recognition by 5–14 months [4, 8, 28]. Within two years, 68% of these patients developed renal disease, highlighting the importance of early detection [1, 6]. Destructive sinonasal lesions were strongly associated with systemic severity. Felicetti et al. observed a 2.8-fold higher risk of glomerulonephritis in patients with destructive lesions (RR 2.8; 95% CI 1.6–4.9; $p < 0.01$) [6]. Cleary et al. reported PR3-ANCA positivity in 88% of patients with bony erosion, supporting a direct association between localized and systemic inflammation [13]. Nasal biopsies confirmed necrotizing granulomas in 94% of cases thus proving ENT sampling as a practical first-line diagnostic approach [2, 3]. PR3-ANCA-driven neutrophil activation explains the cartilage and bone destruction seen in sinonasal disease therefore indicating parallel microvascular injury in renal tissue [2, 3, 5]. Goldfarb et al. also emphasized that ENT manifestations in small-vessel vasculitides, including sinonasal involvement, often precede systemic findings and can aid early diagnosis [29].

Oral disease was less common but highly specific. Strawberry gingivitis was seen in 12–15% of patients, consistently associated with PR3-ANCA positivity and biopsy-proven vasculitis [9, 21, 24]. Dhalkari et al. [24] and Thompson et al. [21] found ANCA positivity in all affected patients. The gingival changes typically appeared 3.5 months before systemic diagnosis [18, 20]. Many cases were initially treated as dental or infectious pathology, delaying therapy [19, 22]. Hanisch et al. [19] identified gingival hyperplasia as a marker of relapse, reinforcing its value in disease monitoring. The dense vascular network of the gingiva and tendency to form neutrophil microabscesses account for its diagnostic reliability [24]. Early recognition of oral lesions, like sinonasal findings, enables biopsy confirmation and faster treatment. It has been reported that gingival swelling was the initial manifestation of GPA, hence signifying its diagnostic potential [30].

Subglottic stenosis was one of the most serious airway complications reported in GPA. Felicetti et al. found it in 21% of adults, with a strong correlation to renal disease activity ($p = 0.007$); 80% of patients with stenosis later developed glomerulonephritis [6]. Pediatric cohorts showed more severe airway involvement. Górný et al. reported

subglottic or tracheal stenosis in 67% (10/15) of children, with several cases requiring tracheostomy due to diagnostic delays [10]. Persistent granulomatous inflammation of the subglottic mucosa caused fibroinflammatory scarring and progressive narrowing. In children, smaller airway calibers made even mild scarring critical, explaining the high tracheostomy rates and greater severity seen in pediatric series [6, 8, 10]. Hegazy et al. [31] also noted that ENT manifestations, including subglottic involvement, can serve as early warning signs for systemic disease progression.

Otologic disease followed a distinct age-related pattern. Adults primarily developed hearing loss or otitis media with effusion, affecting 20–30% of patients. Children more often presented with otologic symptoms as their first sign of GPA, reported in 60% of cases. Sahyouni et al. [11] confirmed GPA through middle ear biopsies in seven patients with chronic otitis. Amiraraghi et al. found a mean diagnostic delay of eight months in chronic otitis cases [12]. Nakamaru et al. reported mastoid involvement in 18%, supporting the use of imaging when bony erosion is suspected [32]. Across studies, ENT manifestations occurred in over 70% of patients, otologic involvement in 25–40%, and serous otitis in 40–70% of ear-related cases [2, 14]. These findings show that nasopharyngeal inflammation and Eustachian tube dysfunction drive persistent effusion, while granulomatous infiltration and vasculitic ischemia disrupt middle ear function. Mastoid bone disease adds diagnostic complexity, often requiring imaging and biopsy [11, 12, 32].

Renal disease occurred in 58% of patients, usually as hematuria or proteinuria. ENT findings preceded renal involvement by about six months in 68% of cases [1, 6]. Rituximab induced remission in 85% of refractory ENT cases, reducing corticosteroid exposure and slowing renal progression [14]. Trimarchi et al. found diagnostic delays tripled the risk of end-stage renal disease [2]. Recognizing ENT disease early offers a primary window to prevent irreversible renal injury [1, 2, 6].

ANCA testing remained central to GPA diagnosis, though results showed variability. PR3-ANCA was positive in 76% of all cases, with studies such as Maqbool et al. reporting 100% positivity in patients with destructive ENT disease [12, 20, 23, 26]. In contrast, D’Onofrio et al. [27] found only 21% ANCA positivity in patients with EGPA-like presentations, showing that seronegative cases and evolving disease phenotypes complicate diagnosis. Maqbool et al. [26] highlighted that repeat ANCA testing confirmed GPA in 40% of initially negative but clinically suspicious cases. These findings demonstrate that while PR3-ANCA strongly correlates with destructive ENT phenotypes, atypical or EGPA-like presentations may remain negative early, delaying recognition. Repeated testing in high-suspicion cases is necessary to avoid missed or late diagnoses.

Table 3 Quality assessment of included case reports and case series using the methodological evaluation tool, with scores for selection, reporting, and overall study quality

S. No	Author (year)	Selection	A1	A2	C1	C2	C3	C4	Reporting	Total score	Quality
1	Amirraghi et al., 2015 [12]	1	1	1	0	1	0	1	1	6/8	Moderate-High
2	Sung IY et al. (2015) [18]	0	1	1	1	0	0	1	1	5/8	Moderate
3	Hanisch M et al. (2016) [19]	1	1	1	0	1	1	0	1	6/8	Moderate-High
4	Fonseca FP (2021) [20]	0	1	1	1	0	0	1	1	5/8	Moderate
5	Kühn et al., 2018 [7]	1	1	1	0	1	1	0	1	6/8	Moderate-High
6	Wojciechowska et al., 2018 [8]	1	1	0	1	0	1	1	1	6/8	Moderate-High
7	Felicetti et al., 2018 [6]	1	1	1	1	1	0	1	1	7/8	High
8	Thompson G et al. (2018) [21]	1	1	0	1	0	1	1	1	6/8	Moderate-High
9	Sahyouni et al., 2019 [11]	1	1	1	1	1	0	1	1	7/8	High
10	Carnevale et al., 2019 [4]	0	1	1	0	0	1	1	1	5/8	Moderate
11	Patrick A et al. (2019) [22]	1	1	1	0	1	1	0	1	6/8	Moderate-High
12	Szczklik K et al. (2019) [23]	1	1	1	1	0	1	1	1	7/8	High
13	Dhalkari CD et al. (2020) [24]	1	1	1	1	1	0	1	1	7/8	High
14	Nico MMS et al. (2020) [25]	1	1	1	0	1	0	1	1	6/8	Moderate-High
15	Maqbool et al., 2023 [26]	1	1	1	1	0	1	1	1	7/8	High
16	D'Onofrio M et al. (2023) [27]	1	0	1	1	0	1	1	1	6/8	Moderate-High
17	Górny et al., 2025 [10]	1	1	1	1	0	1	1	1	7/8	High

Biopsy of accessible ENT sites proved highly effective for early confirmation. Histopathological analysis of nasal, gingival, or subglottic tissue confirmed GPA in 94% of cases, reducing the need for invasive lung or renal biopsies. ENT biopsies shortened diagnostic time by 3.2 months, improving treatment initiation [1]. ENT mucosa is easily accessible, and granulomatous inflammation and vasculitis often appear early, explaining the high diagnostic yield [1–3]. Early biopsy reduces delays, minimizes invasive testing, and accelerates immunosuppression, which prevents irreversible organ damage. Pediatric presentations were distinct. Children showed higher rates of ear and airway involvement and lower rates of nasal deformities (10% compared to 30–50% in adults) [10, 11]. Górny et al. [10] associated these differences to more aggressive local inflammation and longer diagnostic delays (nine months in children versus five in adults), which increased airway narrowing and permanent hearing loss. Narrower airways and higher inflammatory burden explain the greater functional impairment in pediatric GPA thus emphasizing the need for earlier screening.

Quality assessment confirmed reliability. Fourteen of seventeen studies scored $\geq 6/8$ on Murad's tool, and JBI ratings classified studies by Sahyouni et al. [11] and Felicetti et al. [6] as low risk. The Newcastle–Ottawa Scale (NOS) and QUADAS-2 are not suitable for case reports or case series, as they are designed for cohort/case–control and diagnostic accuracy studies, respectively [33, 34]. This review included only 17 case reports and series, for which the JBI checklist and Murad et al. tool is recommended and validated [16, 17]. Meta-analysis was not possible due to small, heterogeneous samples, inconsistent outcome definitions, and variable reporting of renal involvement [35, 36], thus a descriptive synthesis with extracted percentages and key cohort estimates was used [37]. Small sample sizes and retrospective designs limited generalization, but diagnostic consistency across studies remained strong. ENT warning signs destructive sinonasal disease, strawberry gingivitis, subglottic stenosis, and chronic otitis correlated with renal disease risk. Early biopsy and targeted ANCA testing improved diagnostic accuracy and treatment timing. Rituximab induced remission in refractory ENT disease, reduced steroid exposure, and slowed renal decline, supporting its role in treatment algorithms. Recent advances in targeted immunotherapy are changing the treatment options for GPA. Rituximab and standard immunosuppressants are the core therapeutic approaches, but newer agents have also been investigated. Avacopan, a C5a receptor inhibitor, blocks the ANCA-controlled neutrophil activation and has shown non-inferiority to high-dose glucocorticoids with reduced steroid-related toxicity [5]. Mepolizumab, an anti-IL-5 monoclonal antibody, is established in Eosinophilic

Author (year)	Item 1	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7	Item 8	Item 9	Item 10	Overall
Amiraraghi et al., 2015	++	+	++	++	NA	-	++	+	++	+	+
Sung IY et al. (2015)	++	++	++	+	++	++	++	-	++	+	+
Hanisch M et al. (2016)	+	++	++	+	++	++	++	++	-	+	+
Fonseca FP (2021)	++	+	++	++	++	-	++	+	+	++	++
Kühn et al., 2018	++	++	++	++	-	++	++	++	++	++	++
Wojciechowska et al., 2018	++	++	+	NA	-	++	++	++	-	++	+
Felicetti et al., 2018	+	++	++	++	++	++	-	++	++	++	++
Thompson G et al. (2018)	++	+	++	++	++	-	++	+	++	++	++
Sahyouni et al., 2019	++	++	++	++	++	++	-	NA	++	++	++
Carnevale et al., 2019	+	++	++	+	++	++	++	-	++	+	+
Patrick A et al. (2019)	++	+	++	++	++	NA	++	+	-	++	++
Szczklik K et al. (2019)	++	++	+	++	++	++	++	++	++	++	++
Dhalkari CD et al. (2020)	++	++	++	++	++	-	++	+	++	++	++
Nico MMS et al. (2020)	++	+	++	++	++	++	NA	++	NA	++	++
Maqbool et al., 2023	++	+	+	++	++	++	-	++	+	NA	+
D'Onofrio M et al. (2023)	++	++	++	+	++	++	++	-	++	+	+
Gómy et al., (2025)	++	++	+	++	++	++	++	++	++	++	++

Fig. 2 Risk of bias assessment of individual case reports and case series using the JBI tool

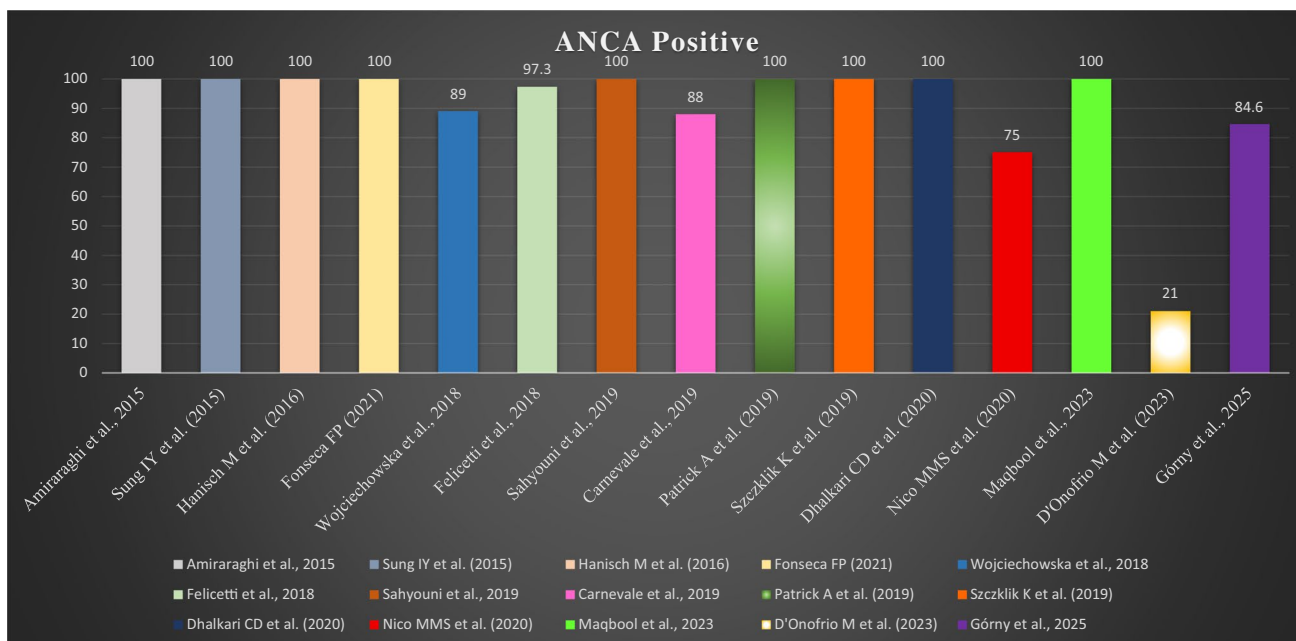


Fig. 3 Number of ANCA-positive cases reported across included studies

Granulomatosis with Polyangiitis (EGPA) and may benefit GPA cases with marked eosinophilic inflammation [27]. Other approaches under evaluation for difficult-to-treat disease include plasma exchange for severe renal involvement, B-cell Activating Factor/B-Lymphocyte Stimulator (BAFF/BlyS) inhibition with agents like belimumab, and Janus Kinase (JAK) inhibitors [15]. These advances highlight the need for accurate phenotyping, including ENT features and ANCA status, to guide individualized therapy and limit long-term organ injury.

Strengths and limitations

A major strength of this systematic review is the comprehensive synthesis of data from 17 studies, including rare case reports and cohort analyses, enabling detailed characterization of otolaryngologic manifestations and their renal correlation in GPA. The use of standardized quality assessment tools ensured methodological rigor. Studies were sourced from diverse geographic regions, enhancing generalizability. However, limitations exist. Most included studies were

retrospective or single-case reports with small sample sizes, increasing risk of selection bias. Heterogeneity in reporting ANCA subtypes and disease activity metrics limited quantitative pooling. Pediatric data were sparse, and long-term outcomes were inconsistently documented. These factors may affect the precision and applicability of the observed associations. The included studies varied in patient populations, including differences in age and adult versus pediatric cohorts. Diagnostic criteria for GPA were inconsistent, with some using ACR 1990, others Chapel Hill 2012, or clinical–serologic–histopathologic consensus. Definitions of renal involvement also differed, ranging from proteinuria to biopsy-confirmed glomerulonephritis. Reporting of ANCA subtypes (PR3 vs. MPO), disease activity measures, and follow-up durations was inconsistent, preventing formal meta-analysis and reducing the accuracy of pooled estimates. Most studies were retrospective or single-case reports with small sample sizes, which increases the risk of selection and reporting bias. Data on pediatric patients were limited, and long-term renal or ENT outcomes were often not reported. These decreases the certainty of the observed associations and emphasizes the need for prospective studies with standardized methods to clarify ENT–renal correlations in GPA.

Conclusion

This review confirms that otolaryngologic manifestations are pivotal in the early diagnosis and risk stratification of GPA, with destructive lesions strongly predicting renal involvement. Clinicians must recognize signs like septal perforation, strawberry gingivitis, and refractory otitis media as red flags warranting ANCA testing and biopsy. Future research should establish standardized ENT disease activity scores and prospective registries to evaluate early intervention outcomes. Longitudinal studies are needed to determine whether timely ENT-directed therapy prevents renal progression, especially in children. Multidisciplinary collaboration between otolaryngologists, rheumatologists, and nephrologists is essential for improving diagnostic speed and treatment efficacy. Developing clinical algorithms based on ENT–renal correlations could enhance patient outcomes. Ultimately, integrating otolaryngologic findings into disease monitoring protocols may reduce morbidity and improve long-term prognosis in GPA.

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